Discrete Mathematics Midterm Exam Collection

Detailed Solutions

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1 Solutions to the 2022 Midterm Exam

1. The quantified definition is

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \left(0 < |x - x_0| < \delta \to |f(x) - A| < \varepsilon\right).$$

2. The statement is true. After removing the four corners, tile the first row except the two corners by three horizontal dominoes, tile the last row the same way, and tile each of the six middle rows by four horizontal dominoes. This covers all remaining 60 squares.

3. They are not logically equivalent. For example, let p be false, q true, and r true. Then

$$p \to (q \to r)$$

is true because p is false, while

$$q \to (p \wedge r)$$

is false because q is true but $p \wedge r$ is false.

4. If $A = B$, then clearly $A \times B = B \times A$. Conversely, suppose $A \times B = B \times A$ and A, B are nonempty. Choose $b_0 \in B$. For any $a \in A$, $(a, b_0) \in A \times B = B \times A$, so $a \in B$. Hence $A \subseteq B$. Similarly, choosing $a_0 \in A$, for any $b \in B$, $(b, a_0) \in B \times A = A \times B$, so $b \in A$. Hence $B \subseteq A$, and therefore $A = B$.

5. (1) If $y \in f(S \cup T)$, then $y = f(x)$ for some $x \in S \cup T$. Thus $x \in S$ or $x \in T$, so $y \in f(S)$ or $y \in f(T)$. Hence $f(S \cup T) \subseteq f(S) \cup f(T)$. The reverse inclusion follows because $S, T \subseteq S \cup T$. Therefore equality holds.

(2) If $y \in f(S \cap T)$, then $y = f(x)$ for some $x \in S \cap T$. Thus $x \in S$ and $x \in T$, so $y \in f(S) \cap f(T)$. Hence $f(S \cap T) \subseteq f(S) \cap f(T)$. Equality need not hold unless f is one-to-one.

6. Since A is countable, list it as $a_1, a_2, \ldots$ if infinite, or as a finite list if finite. Because f is onto, every $b \in B$ occurs as $f(a_i)$ for at least one i. List the elements of B by scanning $f(a_1), f(a_2), \ldots$ and writing down a value only the first time it appears. This gives a finite or countable listing of B, so B is countable.

7. The source file provided for this paper does not include the referenced flowchart, so the exact numbers of additions and multiplications cannot be determined from the available material. Once the flowchart is available, count each addition and multiplication executed in each loop iteration and sum over the loop bounds.

8. For $1 \le j \le n$, we have $j^k \le n^k$. Hence

$$1^k + 2^k + \cdots + n^k \le n \cdot n^k = n^{k+1},$$

so the sum is $O(n^{k+1})$.

9. Since $123 \equiv 22 \pmod{101}$ and 101 is prime, Fermat's little theorem gives $22^{100} \equiv 1 \pmod{101}$. Thus

$$123^{1001} \equiv 22^{1001} = 22(22^{100})^{10} \equiv 22 \pmod{101}.$$

10. The moduli have least common multiple 30. Checking residues modulo 30, $x = 23$ satisfies

$$23 \equiv 5 \pmod{6}, \qquad 23 \equiv 3 \pmod{10}, \qquad 23 \equiv 8 \pmod{15}.$$

Therefore the full solution is

$$x \equiv 23 \pmod{30}.$$

11. Base cases:

$$18 = 7 + 7 + 4, \quad 19 = 7 + 4 + 4 + 4, \quad 20 = 4 + 4 + 4 + 4 + 4, \quad 21 = 7 + 7 + 7.$$

For the induction step, assume every amount from 18 through n can be formed, where $n \geq 21$. Then $n + 1 - 4 \geq 18$, so $n + 1 - 4$ can be formed by the induction hypothesis. Add one 4-cent stamp to form $n + 1$. Thus every amount at least 18 can be formed.

12. (a) A binary tree is either empty, or consists of a root together with a left binary tree and a right binary tree.

(b) One traversal algorithm is preorder traversal:

```
Preorder(T):
    if T is empty:
        return
    visit root(T)
    Preorder(left(T))
    Preorder(right(T))
```

(c) Correctness follows by structural induction. The empty tree has no nodes, so the algorithm visits every node. For a nonempty tree, the algorithm visits the root, then by the induction hypothesis visits every node in the left subtree and every node in the right subtree. These are exactly all nodes of the tree.

2 Solutions to the 2024 Midterm Exam (WYL)

1. The answers are:

$$(1)\ T,\ (2)\ F,\ (3)\ T,\ (4)\ T,\ (5)\ T,\ (6)\ F,\ (7)\ T,\ (8)\ T,\ (9)\ F,\ (10)\ F,\ (11)\ F.$$

For instance, (10) is false because $\sum_{j=1}^{n} j^3$ has order n^4, and (11) is false because $R(3,3) = 6$, not 5.

2. (1) The formula is false only when p is true and q is false. Hence the full DNF is

$$(\neg p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r).$$

(2)
$$P(\{\emptyset, \{\emptyset\}\}) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}.$$

(3)
$$f \circ g = \{(1,7), (2,10), (3,10), (4,7)\}.$$

(4) There are $2^{2^3} = 256$ truth tables.

(5)
$$(\log n)^3, \quad n \log n, \quad n^{99} + n^{98}, \quad n^{100}, \quad 2.5^n, \quad (n!)^2.$$

(6) The numbers are
$$24 \cdot 23 \cdots 5 = \frac{24!}{4!}, \qquad 0.$$

(7)
$$\text{(a) } 4^{10}, \qquad \text{(b) } \binom{13}{3}, \qquad \text{(c) } \binom{9}{3}.$$

(8) The answer is
$$7! + 6 \cdot \frac{7!}{2} = 4 \cdot 7! = 20160.$$

(9) 3256147.

(10) $\{1, 2, 3, 6\}$.

(11) The number is
$$\binom{n+1}{7}.$$
In general, bit strings of length n with exactly k occurrences of the transition 01 are counted by $\binom{n+1}{2k+1}$.

(12) Let $Y(n)$ be the total number, $C(n)$ those ending in a consonant, and $V(n)$ those ending in a vowel. Then $C(n) = 21Y(n-1)$ and $V(n) = 5C(n-1) = 105Y(n-2)$. Thus
$$Y(n) = 21Y(n-1) + 105Y(n-2), \qquad Y(0) = 1, \quad Y(1) = 26.$$

(13) A particular solution has the form
$$An + B + Cn2^n.$$

3. No such onto function exists. If $f : S \to P(S)$ were onto, define

$$T = \{s \in S \mid s \notin f(s)\}.$$

Since $T \subseteq S$, onto-ness gives $T = f(t)$ for some $t \in S$. Then

$$t \in T \iff t \notin f(t) = T,$$

a contradiction. Hence no surjection $S \to P(S)$ exists.

4. The formal form is

$$\big[(p \vee q) \wedge (\neg r \to \neg p) \wedge (r \to s) \wedge \neg q\big] \to s.$$

The argument is valid: from $p \vee q$ and $\neg q$, infer p; from $\neg r \to \neg p$, use contraposition to get $p \to r$; hence r; then from $r \to s$, infer s.

5. Partition $\{1, 2, \ldots, 2n\}$ into the n pairs

$$\{1, 2\}, \{3, 4\}, \ldots, \{2n - 1, 2n\}.$$

Selecting $n + 1$ integers forces two selected integers to lie in the same pair. Consecutive integers are relatively prime, so those two selected integers are relatively prime.

6. Count $(r + 1)$-element subsets of $\{1, 2, \ldots, n + r + 1\}$ by their largest element. If the largest element is $n + k + 1$, where $0 \le k \le r$, then the remaining r elements are chosen from $n + k$ elements, giving $\binom{n+k}{r} = \binom{n+k}{k}$ choices. Summing over k gives the left side, while direct counting gives the right side.

7. Let a_n be the number of ways and set $a_0 = 1$, $a_n = 0$ for $n < 0$. The last item entered can be one of two \$1 items, one \$2 item, or one of two \$5 items. Therefore

$$a_n = 2a_{n-1} + a_{n-2} + 2a_{n-5} \qquad (n \ge 1),$$

with $a_0 = 1$ and $a_n = 0$ for $n < 0$.

3 Solutions to the 2024 Midterm Exam (LLF)

1.

$$(1)\ T, \quad (2)\ F, \quad (3)\ F, \quad (4)\ T, \quad (5)\ T.$$

The cancellation in (2) needs $\gcd(c, m) = 1$, and $\mathbb{Z}$ is not well-ordered under the usual order because it has nonempty subsets with no least element.

2. (1) The characteristic equation of the homogeneous part is $(r-2)^2 = 0$. Since the forcing term is a degree-2 polynomial times 2^n, a particular solution has the form

$$a_n^{(p)} = n^2 2^n (An^2 + Bn + C).$$

(2) In `multiplication`, the letter i appears 3 times, l appears 2 times, and t appears 2 times. Hence the number is

$$\frac{14!}{3!\,2!\,2!}.$$

(3) 0, because $|A| > |B|$.

(4) There are $10 \cdot 9 \cdot 8 = 720$ possible codes. To force six identical codes, the minimum is

$$5 \cdot 720 + 1 = 3601.$$

(5) Put $y_i = x_i - 2$. Then $y_1 + \cdots + y_5 = 11$, so the number is

$$\binom{15}{4}.$$

(6)

$$\frac{x^3}{(1+4x)^2} = x^3 \sum_{n \geq 0} (-1)^n (n+1) 4^n x^n.$$

The coefficient of x^{12} is the coefficient for $n = 9$:

$$-10 \cdot 4^9.$$

(7) By inclusion-exclusion, the number of onto assignments of 9 distinct reports to 4 employees is

$$4^9 - \binom{4}{1} 3^9 + \binom{4}{2} 2^9 - \binom{4}{3} 1^9 = 186480.$$

(8) Since 23 is prime and $9^{22} \equiv 1 \pmod{23}$,

$$9^{45} = 9^{44} 9 \equiv 9 \pmod{23}.$$

(9) Euclid's algorithm gives

$$450 = 3 \cdot 120 + 90, \qquad 120 = 90 + 30.$$

Therefore

$$30 = 120 - 90 = 120 - (450 - 3 \cdot 120) = 4 \cdot 120 - 450.$$

(10) For each diagonal pair there are 2 choices, and for each unordered pair of distinct elements there are 3 choices: neither ordered pair, the first direction, or the second direction. Since there are $\binom{4}{2} = 6$ unordered distinct pairs, the answer is

$$2^4 3^6.$$

3. (1) Base: $4 \equiv 4 \pmod 6$. If $n \equiv 4 \pmod 6$, then

$$5n + 2 \equiv 5 \cdot 4 + 2 = 22 \equiv 4 \pmod 6,$$

and

$$n^2 \equiv 4^2 = 16 \equiv 4 \pmod 6.$$

Thus every generated element is congruent to 4 modulo 6.

(2) The integer 10 satisfies $10 \equiv 4 \pmod 6$ but $10 \notin S$: the only basis element is 4, and every recursive operation applied to a positive element of S produces a number at least 16.

4. Let $G(x) = \sum_{n \geq 0} a_n x^n$. From the recurrence,

$$G(x) - 1 - x = 7x(G(x) - 1) - 10x^2 G(x),$$

so

$$G(x) = \frac{1 - 6x}{1 - 7x + 10x^2} = -\frac{1}{3} \cdot \frac{1}{1 - 5x} + \frac{4}{3} \cdot \frac{1}{1 - 2x}.$$

Therefore

$$a_n = \frac{4 \cdot 2^n - 5^n}{3}.$$

5. The generating function for the choices is

$$\left(1 + x^2 + x^4 + \cdots + x^{2\lfloor r/2 \rfloor}\right)(1 + x + \cdots + x^r)^2.$$

The answer is the coefficient of x^{12} in this product. If $r \geq 12$, the upper bounds do not matter for a total of 12, and the count is

$$\sum_{j=0}^{6} (13 - 2j) = 49.$$

6. (1) The last coin is either \$1 or \$2, so

$$a_n = a_{n-1} + a_{n-2}, \qquad a_0 = 1, \quad a_1 = 1.$$

(2) This is the Fibonacci recurrence. With $\varphi = (1 + \sqrt{5})/2$ and $\psi = (1 - \sqrt{5})/2$,

$$a_n = \frac{\varphi^{n+1} - \psi^{n+1}}{\sqrt{5}}.$$

4 Solutions to the 2025 Midterm Exam (WYL)

1.

$$(1)\ T,\ (2)\ T,\ (3)\ F,\ (4)\ T,\ (5)\ F,\ (6)\ F,\ (7)\ F,\ (8)\ F,\ (9)\ T,\ (10)\ F.$$

2. (1)

$$\forall A \forall B \big((S(A,B) \wedge F(B)) \to F(A) \big).$$

(2) Standardize variables first:

$$(\exists u P(u) \wedge \forall v Q(v)) \to \exists w (P(w) \vee Q(w)).$$

This is equivalent to

$$\forall u \exists v \exists w \left(\neg P(u) \vee \neg Q(v) \vee P(w) \vee Q(w) \right).$$

(3)

$$\{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}.$$

(4)

$$f \circ g = \{(1,7), (2,10), (3,3), (4,7)\}.$$

(5) $2^{2^2} = 16$.

(6)

$$(\log n)^3, \quad \sqrt{n} \log n, \quad n^{99} + n^{98}, \quad n^{100}, \quad (1.5)^n, \quad 10^n, \quad (n!)^2.$$

(7)

$$12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = \frac{12!}{6!}, \qquad 0.$$

(8)

$$\text{(a) } 10^{19}, \qquad \text{(b) } \binom{28}{9}, \qquad \text{(c) } \binom{18}{9}.$$

(9)

$$\frac{8!}{3!\,2!} = 3360.$$

(10) 31542678.

(11)

$$\{2,3,5,6\}, \quad \{2,3,5,7\}, \quad \{2,3,5,8\}, \quad \{2,3,6,7\}.$$

(12) $\binom{n+1}{5}$.

(13) Each element of S has three choices: in neither set, in B only, or in both A and B. Hence the answer is $3^2 = 9$.

3. The formal form is

$$[\forall x (M(x) \vee F(x)) \wedge \forall x (M(x) \leftrightarrow G(x)) \wedge \neg \forall x G(x)] \to \exists x F(x).$$

The argument is valid. From $\neg \forall x G(x)$, get $\exists x \neg G(x)$. For such an x, $M(x) \leftrightarrow G(x)$ gives $\neg M(x)$. Since $M(x) \vee F(x)$, it follows that $F(x)$.

4. If f is one-to-one, then its restriction to any subset $S \subseteq A$ is also one-to-one, so the finite sets S and $f(S)$ have the same size. Conversely, suppose $|f(S)| = |S|$ for all $S \subseteq A$. If f were not one-to-one, there would be $x \neq y$ with $f(x) = f(y)$. For $S = \{x, y\}$, we would have $|S| = 2$ but $|f(S)| = 1$, a contradiction. Hence f is one-to-one.

5. Suppose, for contradiction, that some integer greater than 1 has no prime divisor. By well-ordering, let n be the least such integer. Then n is not prime, so $n = ab$ with $1 < a < n$ and $1 < b < n$. By minimality of n, the integer a has a prime divisor p. Since $p \mid a$ and $a \mid n$, we have $p \mid n$, contradicting the choice of n.

6. A bit string of length at most 9 is determined, for this problem, by the pair (z, o) giving its number of 0s and 1s, with $z, o \geq 0$ and $z + o \leq 9$. The number of possible pairs is

$$\sum_{\ell=0}^{9} (\ell + 1) = 55.$$

Since there are 101 strings and only 55 possible pairs, the pigeonhole principle gives two strings with the same numbers of 0s and 1s.

7. Both sides count ordered $(r+1)$-tuples of distinct elements chosen from an $(n+1)$-element set. The left side first chooses an ordered r-tuple and then one more element from the remaining $n + 1 - r$ elements. The right side first chooses the first element, then chooses an ordered r-tuple from the remaining n elements.

8. Let $N = R(m, n-1) + R(m-1, n)$. In any party of N people, pick one person v. Among the other $N-1$ people, either at least $R(m-1, n)$ are friends of v, or at least $R(m, n-1)$ are enemies of v. In the first case, the friend set contains either $m - 1$ mutual friends, which together with v give m mutual friends, or n mutual enemies. In the second case, the enemy set contains either m mutual friends, or $n - 1$ mutual enemies, which together with v give n mutual enemies. Thus $R(m, n) \leq N$.

5 Solutions to the 2025 Midterm Exam (ZYW)

1. No allowed formula splits the 8 truth assignments into 4 and 4; every allowed read-once formula using $\wedge, \vee, \neg$ is true on $1, 3, 5$, or 7 assignments. Hence three rounds cannot guarantee success: after the first answer, at least five assignments may remain, and two more yes/no answers distinguish at most four cases. Thus at least four rounds are needed.

A four-round strategy is the following decision tree. Ask

$$Q_1 = \neg p \wedge \neg q \wedge \neg r.$$

If true, the assignment is FFF. If false, ask

$$Q_2 = (\neg p \wedge \neg q) \vee \neg r.$$

If Q_2 is true, ask

$$Q_3 = \neg p \wedge (\neg q \vee \neg r).$$

If Q_3 is true, ask $Q_4 = (\neg p \vee \neg q) \wedge \neg r$; true gives FTF, false gives FFT. If Q_3 is false, ask the same Q_4; true gives TFF, false gives TTF.

If Q_2 is false, ask

$$Q_3' = \neg p \vee (\neg q \wedge \neg r).$$

If true, the assignment is FTT. If false, ask

$$Q_4' = \neg p \vee \neg q \vee \neg r.$$

True gives TFT, false gives TTT. Therefore four rounds are sufficient and minimal.

2. Using $(a, b) = \{a, \{a, b\}\}$:

(i)
$$\{\{1, \{1, 2\}\}\}.$$

(ii)
$$\{\{1, \{1, 2\}\}, \{1, \{1, 3\}\}, \{2, \{2\}\}, \{2, \{2, 3\}\}\}.$$

(iii) $\emptyset$.

(iv) Let $u = \{1, \{1, 2\}\}$. Then
$$\{\{u, \{u, 1\}\}, \{u, \{u, 2\}\}\}.$$

(v)
$$\big\{\{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset, \{\emptyset\}\}\}\big\}.$$

3. (i) A standard bijection is
$$f(x) = \tan\big(\pi(x - \tfrac{1}{2})\big),$$
which maps $(0, 1)$ bijectively onto $\mathbb{R}$.

(ii) Use decimal expansions that do not end in all 9s. If

$$x = 0.a_1a_2a_3\cdots, \qquad y = 0.b_1b_2b_3\cdots,$$

define

$$F(x,y) = 0.a_1b_1a_2b_2a_3b_3\cdots.$$

The interleaved decimal determines both original decimals, so F is injective.

4.
```
ExactlyTwice(A[1..n]):
    count[1..1000] = all zeros
    for i = 1 to n:
        count[A[i]] = count[A[i]] + 1
    for v = 1 to 1000:
        if count[v] == 2:
            output v
```

The only comparisons outside loop tests are the 1000 comparisons $count[v] == 2$, so the worst-case number is 1000.

5. Suppose b and c are both inverses of a modulo m. Then

$$ab \equiv 1 \pmod{m}, \qquad ac \equiv 1 \pmod{m}.$$

Hence $ab \equiv ac \pmod{m}$, so $a(b-c) \equiv 0 \pmod{m}$. Since $\gcd(a,m) = 1$, we can cancel a modulo m, giving $b \equiv c \pmod{m}$. Thus the inverse is unique modulo m.

6. Write $a = qb + r$ with $0 \le r < b$. Since $2^b \equiv 1 \pmod{2^b - 1}$,

$$2^a = 2^{qb+r} = (2^b)^q 2^r \equiv 2^r \pmod{2^b - 1}.$$

Therefore

$$2^a - 1 \equiv 2^r - 1 \pmod{2^b - 1},$$

where $r = a \bmod b$.

7. (i) Since $3^6 \equiv 1 \pmod{7}$ and $2003 \equiv 5 \pmod{6}$,

$$3^{2003} \equiv 3^5 = 243 \equiv 5 \pmod{7}.$$

(ii) To find the last digit of $\lfloor 3^{2003}/7 \rfloor$, compute $3^{2003} \pmod{70}$. Since $3^{2003} \equiv 47 \pmod{70}$,

$$3^{2003} = 70q + 47.$$

Then

$$\left\lfloor \frac{3^{2003}}{7} \right\rfloor = 10q + 6,$$

so the last digit is 6.

8. Let T_n be the displayed implication. For $n = 2$, T_2 is

$$(p_1 \to p_2) \to (p_1 \to p_2),$$

a tautology. Assume T_n is a tautology. For $n + 1$, if

$$(p_1 \to p_2) \wedge \cdots \wedge (p_n \to p_{n+1})$$

and $p_1 \wedge \cdots \wedge p_n$ are true, then by the induction hypothesis applied to the first n propositions, p_n is true. Since $p_n \to p_{n+1}$ is true, p_{n+1} is true. Hence the implication for $n + 1$ is true, completing the induction.

9. The amounts are 2 and every integer amount at least 4; 1 and 3 cannot be formed. For the induction proof, show every $n \geq 4$ can be formed. Base cases: $4 = 2 + 2$ and $5 = 5$. For $n \geq 6$, if $n - 2$ can be formed, then adding one two-dollar bill forms n. Strong induction therefore proves all amounts at least 4 can be formed.

10. (i) Iterative version:

```
Iterative(n):
    if n == 0: return 1
    if n == 1: return 3
    if n == 2: return 5
    a0 = 1; a1 = 3; a2 = 5
    for k = 3 to n:
        a3 = a2 * a1^2 * a0^3
        a0 = a1; a1 = a2; a2 = a3
    return a2
```

Recursive version:

```
Recursive(n):
    if n == 0: return 1
    if n == 1: return 3
    if n == 2: return 5
    return Recursive(n-1) * Recursive(n-2)^2 * Recursive(n-3)^3
```

(ii) The iterative algorithm is more efficient. It computes each previous term once, so it uses $O(n)$ recurrence steps, while the naive recursive algorithm recomputes the same terms many times.

6 Solutions to the 2025 Midterm Exam (LLF)

1.

$$(1)\ F, \quad (2)\ T, \quad (3)\ F, \quad (4)\ F, \quad (5)\ T.$$

For (2), $2(-15) = -30 \equiv 1 \pmod{31}$. For (3), the number of unordered bipartitions into two nonempty parts is $2^9 - 1 = 511$, not 512. For (4), the recurrence also applies at $n = 1$, forcing $f(1) = 2f(0) = 0$, which contradicts $f(1) = 1$.

2. (1) 9.

(2) There are 4^{10} possible answer sheets. To force three identical sheets, the minimum is

$$2 \cdot 4^{10} + 1.$$

(3) There are 5^8 functions and 0 one-to-one functions.

(4) The string has 23 characters. The multiplicities are i^5, l^3, e^3, and a, t, c, n each twice. Hence the number is

$$\frac{23!}{5!\,3!\,3!\,(2!)^4}.$$

(5) Put $y_i = x_i - 2$. Then $y_1 + y_2 + y_3 + y_4 = 12$, so the answer is

$$\binom{15}{3}.$$

(6)

$$(1+2x)^{-2} = \sum_{n\geq 0} (-1)^n (n+1) 2^n x^n.$$

The coefficient of x^{12} in $x^3(1+2x)^{-2}$ is the coefficient of x^9 in $(1+2x)^{-2}$:

$$-10 \cdot 2^9 = -5120.$$

(7)

$$4^8 - \binom{4}{1}3^8 + \binom{4}{2}2^8 - \binom{4}{3}1^8 = 40824.$$

(8)

$$a_n = a_{n-1} + a_{n-2}, \qquad a_0 = 1, \quad a_1 = 1.$$

(9) All diagonal pairs must be included. For each unordered pair of distinct elements, either include both directions or include neither. Therefore the number is

$$2^{\binom{n}{2}} = 2^{n(n-1)/2}.$$

3. (1) Starting from $(0, 0)$, the first application gives

$$(2, 3), (3, 2).$$

The second gives

$$(4, 6), (5, 5), (6, 4).$$

The third gives

$$(6, 9), (7, 8), (8, 7), (9, 6).$$

(2) Basis: $(0,0)$ satisfies $5 \mid 0+0$. If $5 \mid a+b$, then

$$(a+2)+(b+3) = a+b+5$$

and

$$(a+3)+(b+2) = a+b+5$$

are also divisible by 5. By structural induction, $5 \mid a+b$ for every $(a,b) \in S$.

4. Euclid's algorithm gives

$$84 = 4 \cdot 18 + 12, \qquad 18 = 12 + 6.$$

Hence

$$6 = 18 - 12 = 18 - (84 - 4 \cdot 18) = 5 \cdot 18 - 84.$$

5. Solve

$$a_n = 3a_{n-1} + 2^n + 5, \qquad a_0 = 1.$$

A homogeneous solution is $C3^n$. Try a particular solution $A2^n + B$. Substitution gives

$$A2^n = 3A2^{n-1} + 2^n, \qquad B = 3B + 5,$$

so $A = -2$ and $B = -5/2$. Thus

$$a_n = C3^n - 2^{n+1} - \frac{5}{2}.$$

From $a_0 = 1$, $C - 2 - \frac{5}{2} = 1$, so $C = \frac{11}{2}$. Therefore

$$a_n = \frac{11}{2}3^n - 2^{n+1} - \frac{5}{2}.$$

6. Index the sequence by $a_0 = 2, a_1 = 5, a_2 = 9, \ldots$.

(1) The successive differences are $3, 4, 5, 6, \ldots$, so

$$a_n = a_{n-1} + n + 2 \qquad (n \geq 1), \quad a_0 = 2.$$

(2) Solving gives

$$a_n = 2 + \sum_{k=1}^{n}(k+2) = 2 + \frac{n(n+1)}{2} + 2n = \frac{n^2+5n+4}{2}.$$

(3) The generating function is

$$G(x) = \sum_{n\geq 0} a_n x^n = \frac{1}{2}\sum_{n\geq 0} n^2 x^n + \frac{5}{2}\sum_{n\geq 0} n x^n + 2\sum_{n\geq 0} x^n.$$

Using

$$\sum nx^n = \frac{x}{(1-x)^2}, \qquad \sum n^2 x^n = \frac{x(1+x)}{(1-x)^3},$$

we get

$$G(x) = \frac{2-x}{(1-x)^3}.$$